

Thien Nguyen

STG-345

02/09/2020

Professor: Kevin William

Benchmark: SolidWorks Intersecting Pipe Network Simulation

Drawing of the Pipe Network

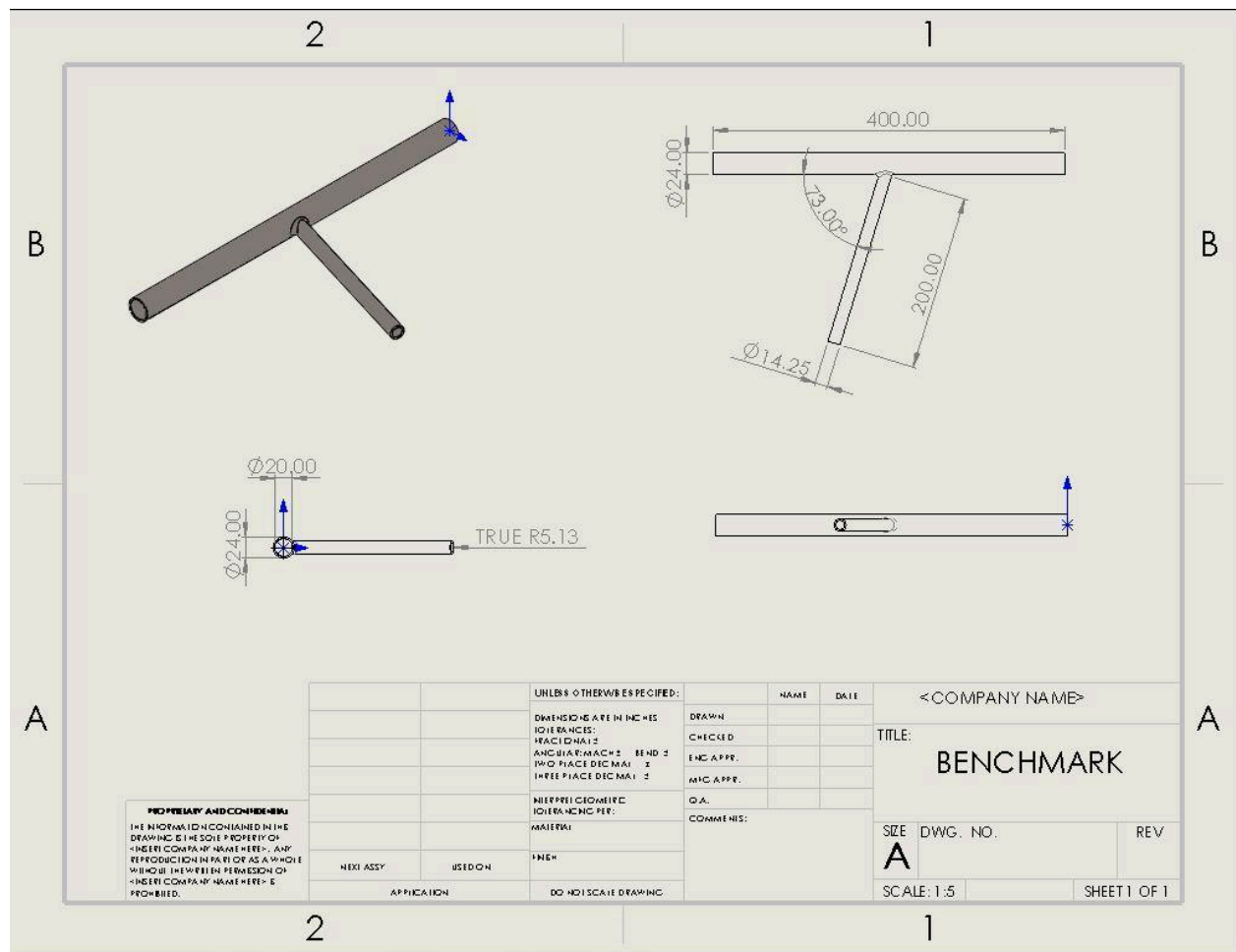


Figure 1: Drawing of the Pipe Network

Figure 1 is the drawing of the pipe network of this model. It is important to note that the pipe has a thickness of 2.0 mm. The main pipe and the intersecting pipe have an inner radius of 20.0 mm and 10.25 mm respectively.

Inlet Mass Flow Rates

Table 1: Raw Data

	D [m]	L [m]	Re	θ [°]	μ [Pa · s]
Main pipe	0.02	0.40	500	73	1.0019×10^{-3}
Intersecting pipe	0.01025	0.20	5000		

Equations used:

$$Q = \frac{w}{\rho}$$

$$v = \frac{Q}{A_c} = \frac{w}{\rho A_c}$$

$$Re = \frac{\rho v D}{\mu}$$

$$w = \frac{Re \mu A_c}{D} = \frac{Re \mu (\pi R^2)}{2R} = \frac{1}{2} \pi Re \mu R$$

Where:

Q: volume flow rate [m³/s]

w: mass flow rate [kg/s]

ρ : density [kg/m³]

v : velocity [m/s]

A_c : cross-sectional area [m²]

Re: Reynolds number

D: diameter [m]

μ : dynamic viscosity [Pa · s]

Mass flow rates calculations:

$$w_m = \frac{1}{2} \pi (500) \left(1.0019 \times 10^{-3} \text{ Pas} \right) \left(\frac{0.02}{2} \text{ m} \right) = 7.87 \frac{\text{g}}{\text{s}}$$

$$w_i = \frac{1}{2} \pi (5000) \left(1.0019 \times 10^{-3} \text{ Pas} \right) \left(\frac{0.01025}{2} \text{ m} \right) = 40.33 \frac{\text{g}}{\text{s}}$$

The main pipe and the intersecting pipe have a mass flow rate of 7.87 g/s and 40.33 g/s according to the Reynolds number for both inflow legs. All raw data is provided in table 1 above.

Laminar and Turbulent Flow at the Inlets

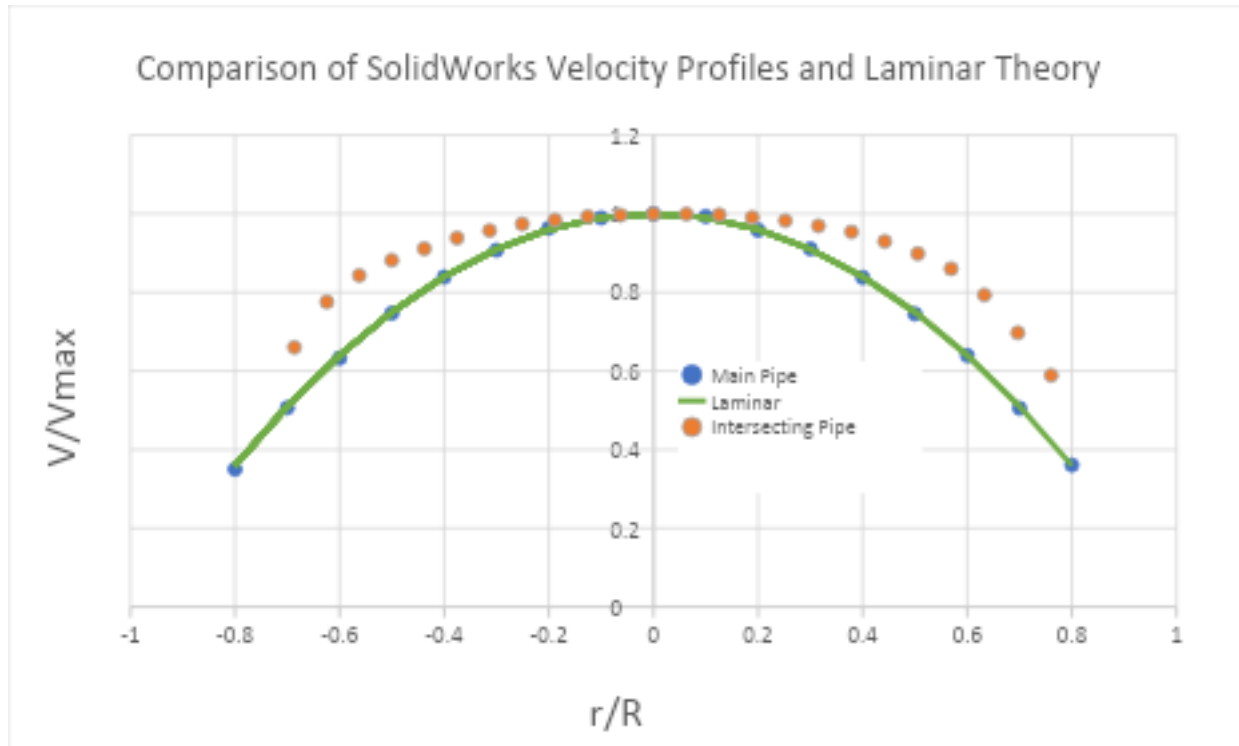


Figure 2: Comparison of SolidWorks Velocity Profiles and Laminar Theory

As shown in figure 2, the laminar theory curve (green) fits the SolidWorks Simulation at the inlet of the main pipe (blue). In addition, the Reynolds number at the inlet of the main pipe is 500 which is less than 2100. It is known that a Reynolds number that is less than 2100 indicates a high possibility of a laminar flow. Therefore, it is reasonable to conclude that the main pipe has a laminar inflow.

The velocity profile at the inlet of the main pipe (blue) is axisymmetric and has a shape of a bell curve. Meanwhile, the velocity profile at the inlet of the intersecting pipe (orange) has a curve of a fully developed turbulent flow. The turbulent curve is wider than the laminar curve.

Both laminar and turbulent flows indicate the highest velocity at the center of the pipes. However, as going from the center to the wall, the laminar flow has a lower velocity than turbulent flow.

Because of the viscosity of water, the centered water molecules (with the highest velocity) drag the particles that are next to them as they move through the pipe. Shear force is then generated creating the momentum diffusion from the center to the pipe's wall. This phenomenon is well-explained by the laminar flow in figure 2. Turbulent flow, as indicated by its name, does not have consistency in its behavior. It is easy to see that almost all of the particles that are not at the wall have a high velocity. The closer the molecules to the wall, the smaller their velocity is. The viscosity between water and the steel wall creates a shear force that prevents the water molecules from moving along the pipe. Diffusion still occurs but the domination of the convective flux creates such a uniform average profile shown in figure 2. In other words, turbulent flow occurs when the inertial forces and the velocity are much larger than the drag force.

The Mixing Process

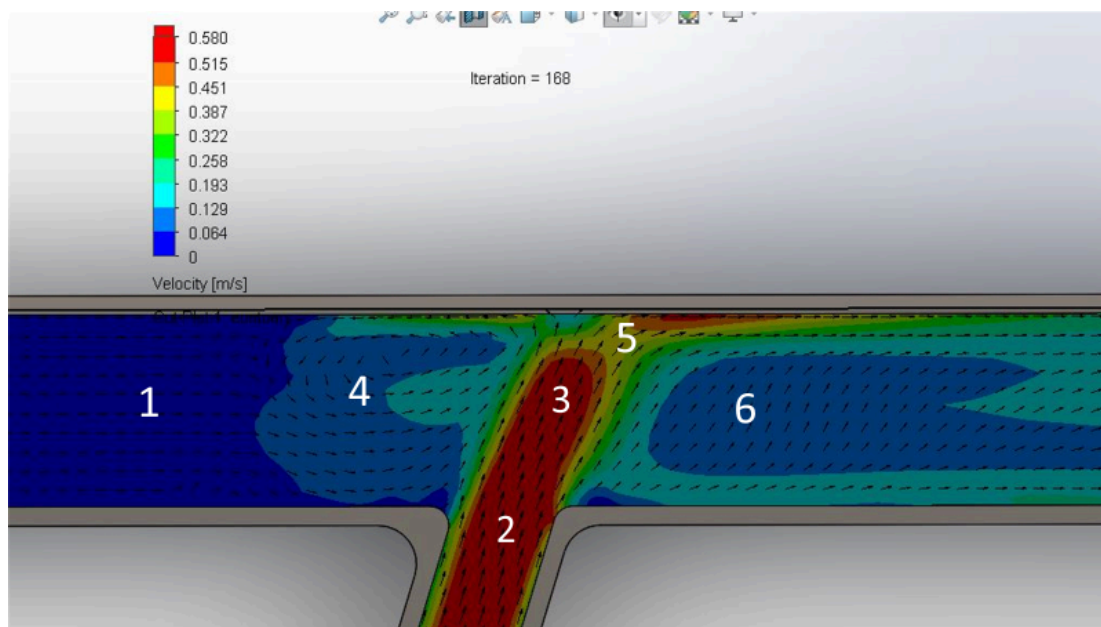


Figure 3: Velocity Profiles at the Mixing Region

For the simplest description, all directions will be indicated as left, right, upward and downward in figure 3. The laminar flow of the main pipe (region 1) is moving the right while the turbulent flow of the intersecting pipe is moving upward and slightly to the right. At the intersection (region 3), the flows meet each other creating different mixing regions 4, 5 and 6.

From regions 2 to 3, as the turbulent flow has a much higher momentum, its particles dominate the molecules of the laminar flow until they hit the wall and separate into two different directions. One of which moves back to the left and the other one continues to flow to the right along the pipe. At region 4, the particles from region 3 that flow back the left meet the coming particles of the laminar moving to the right. As their directions are much opposite, the particles start to circulate themselves into a vortex (region 4).

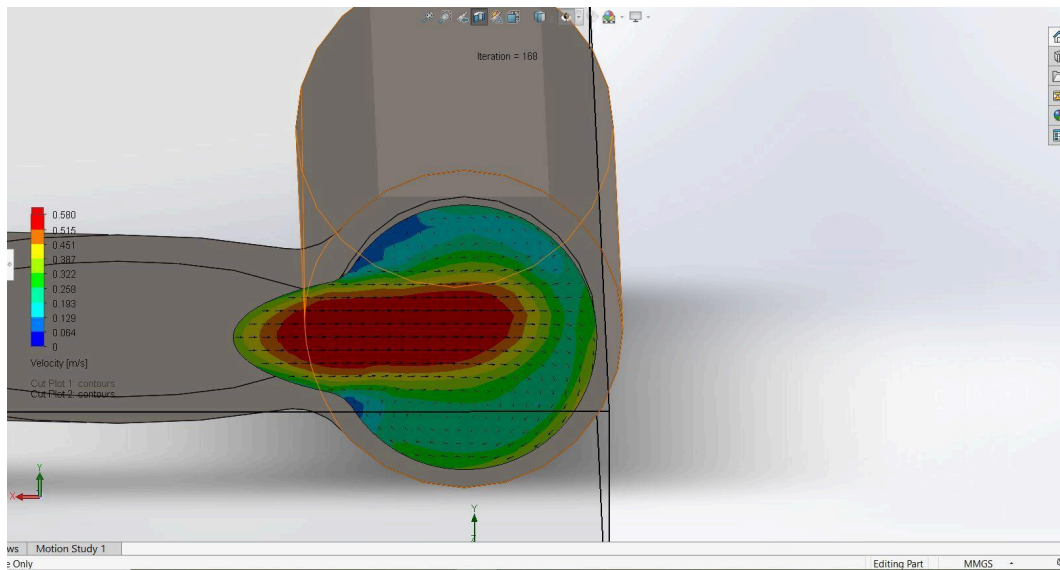


Figure 4: Side View of the Velocity Profiles at the Mixing Region

Meanwhile, the particles from region 3 that continue to move the right after hitting the wall meet with the water molecules of the laminar flow the main pipe snaking along the top and bottom of the main pipe's wall (see figure 4). As these molecules have a similar direction (rightward), they continue to drag each other along the main pipe from the mixing region to the outlet. It is important to note that the water particles of the region 2 and 3 have such a high momentum, they dominate and diffuse their momentum to the particles that are next to them. This explains the upward and rightward velocity profiles of the water molecules in region 6.

The Outflow Condition

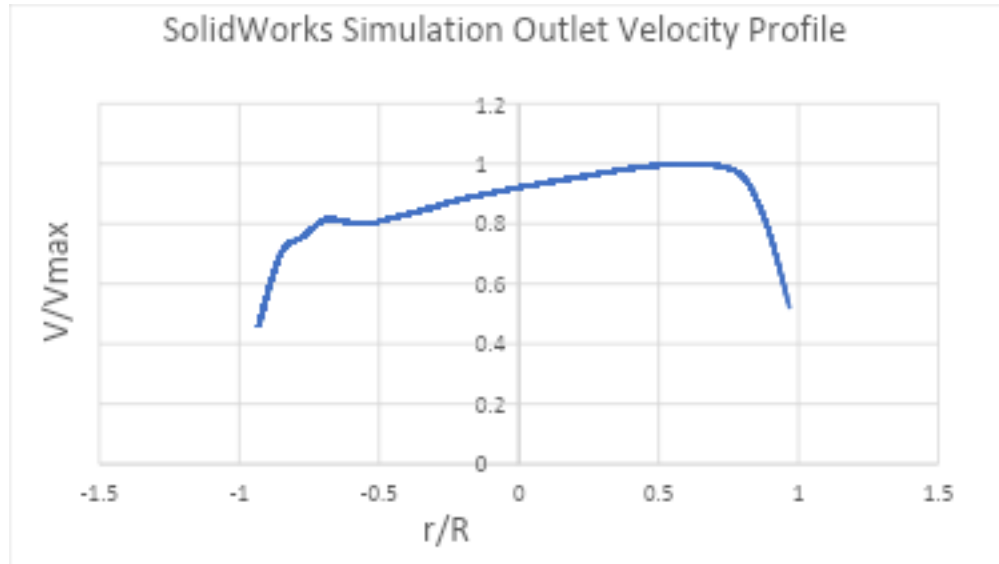


Figure 5: SolidWorks Simulation Outlet Velocity Profile

Homogenization Region

Homogenization does not happen within this model. The outflow chart shows that the flow is not yet fully developed. This means the flow is neither laminar nor turbulent at the outlet. Also, more time and distance are needed for the flow to develop fully. According to figure 5, if the length of the main pipe is tripled, the flow will have a high chance to be fully developed into either laminar or turbulent.

Discussion of the Velocity Profile of the Pipe Network

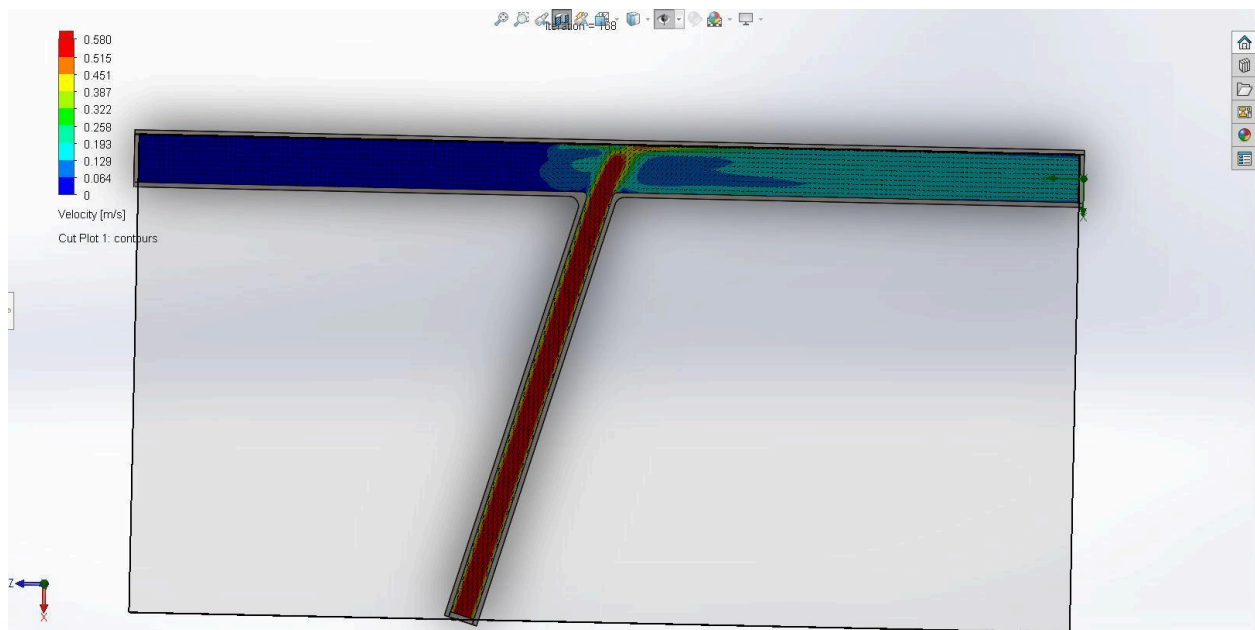


Figure 6: Top View of the Overall Velocity Profile of the Pipe System

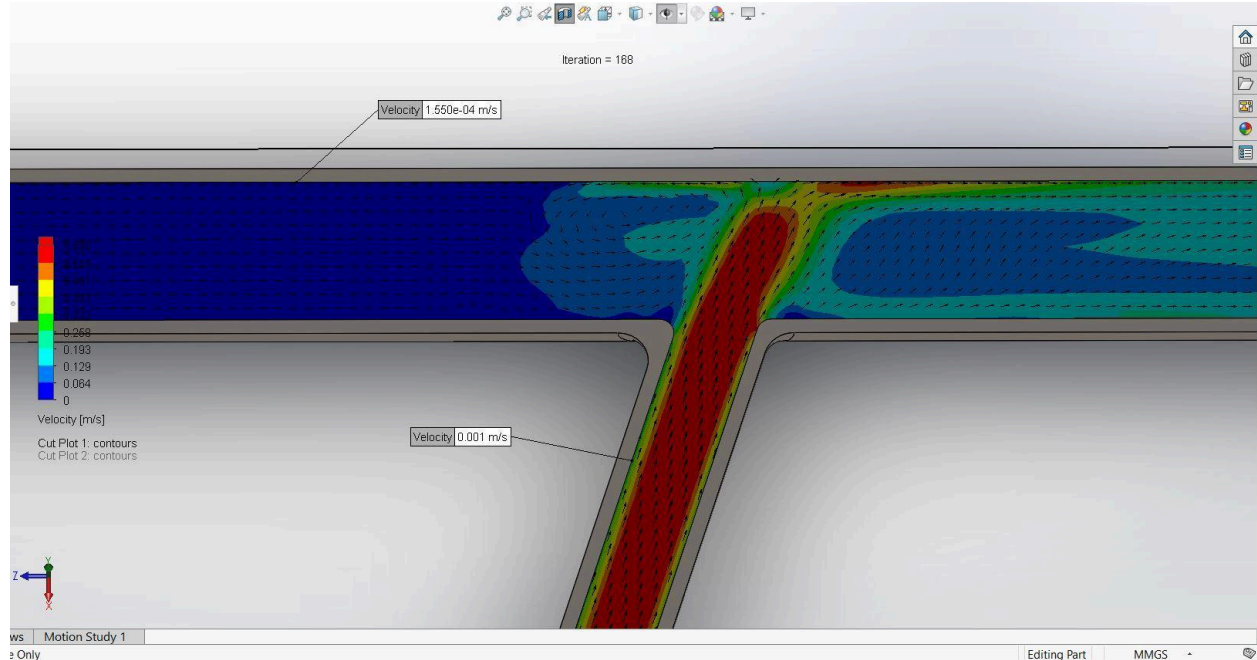


Figure 7: Probe Data of the Velocity at the Wall of the Pipe

It is known that the flow velocity is generally zero [m/s] at the wall throughout the whole pipe. Figure 7 also confirms that this knowledge is true and accurate in this model. From figure 6, SolidWorks demonstrates the water velocity increases from dark blue to red. Therefore, the laminar flow of the main pipe inlet is the region that has the lowest velocity while the turbulent flow of the intersecting pipe inlet is the region that has the highest velocity.

In region 4 (see figure 3), the center of the vortex may also have zero velocity. In other words, this location can be considered as a very small stagnant area in the network. Probing the velocity of the center of the vortex was attempted; however, the velocity was not zero. It could have been because manually probing the center was inaccurate.